

CLINICAL INVESTIGATION

Eye

CONSERVATION TREATMENT OF THE EYE: CONFORMAL PROTON REIRRADIATION FOR RECURRENT UVEAL MELANOMA

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Purpose: To evaluate the outcomes of a second course of proton beam radiation therapy (PBRT) in patients with recurrent uveal melanoma.

Methods and Materials: Thirty-one patients received a second course of PBRT. The mean interval between the first and the second PBRT course was 50.2 months (range, 8–165 months). Most patients (87%) received 70 cobalt Gray equivalent (CGE) for both courses. Visual acuity was 20/200 or better in 30 patients initially and in 22 patients at the second treatment. The mean follow-up time after the second treatment was 50 months (range, 6–164 months).

Results: At the time of the last follow-up, 20 patients were classified as having no evidence of disease, defined as tumor regression or an absence of tumor progression. Nine eyes (29%) were enucleated because of local recurrence ($n = 5$) or intractable pain ($n = 4$). The 5-year eye retention rate was 55% (95% confidence interval: 25.2–77.4). Six of the 22 patients who retained the eye (27%) had useful vision (20/200 or better).

Conclusions: A second course of PBRT for recurrent uveal melanoma to total doses between 118 and 140 CGE was associated with a relatively good probability of local control and a low enucleation rate. Although most patients lost vision, the majority were able to retain the reirradiated eye. Further evaluation is needed to assess metastasis-free survival of additional proton irradiation vs. enucleation after local recurrence. © 2006 Elsevier Inc.

Proton beam radiation therapy, Uveal melanoma, Eye, Survival rate.

INTRODUCTION

The standard treatment for uveal melanoma was enucleation until alternative approaches aimed at eye conservation emerged in the 1960s (1). Excellent results were reported with radiation therapy, particularly with heavy charged particles, such as protons and helium ions (1–9) and with episcleral plaque brachytherapy (1–4, 7, 9–12). Currently, conservation treatment of uveal melanoma is considered standard treatment for many such tumors.

Data recently published on the results of proton treatment at the Harvard Cyclotron Laboratory showed a local control rate at 5 years of approximately 95% and an overall survival rate of 80%, with a relatively low rate of complications (7). Despite these good results, 5% of the patients experienced a relapse in the treated eye and underwent enucleation.

Reirradiation of the eye may be an alternative to enucleation because of the properties of charged particle therapy:

- Proton beam irradiation is a highly conformal treatment that allows the delivery of the dose to the tumor and limits the dose to surrounding critical structures.
- Additionally, the eye is very resistant to focal irradiation, as demonstrated by the use of plaque brachytherapy.

These properties may allow for a second course of therapy without a significant increase in radiation-induced morbidity.

Data published previously suggest that patients treated with proton beam radiation therapy (PBRT) had a 15-year survival rate, similar to those treated with enucleation (13, 14). Starting in 1984, a second course of PBRT was offered to select patients with recurrent melanoma, that is, to those patients who might retain useful vision and were motivated to retain the eye.

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